

MyDesign Portfolio

Element E Initial Template

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

STEM Principle for High-Priority Design Requirement #1

STEM Concept:

HS-ETS1-3 Engineering Design (NGSS)

Describe how STEM concept applies directly to the specifics of your design:

Prior to finalizing a design, each member of our team came up with a possible design solution of their own according to the criteria and constraints. A floor plan of the homes were designed and modeled on the Planner 5D platform. Systems diagram posters were created for the detection/notification and suppression systems. Of those ideas, we chose the design which seemed the most effective in detecting and suppressing fire, was aesthetically pleasing, and was possible to create with the available resources.

STEM Principle for High-Priority Design Requirement #2

STEM Concept:

HS-ETS1-1 Engineering Design (NGSS)

Describe how STEM concept applies directly to the specifics of your design:

The major global challenge which our project addresses is the need for effective fire detection and suppression systems. While performing research, our team found that fire related deaths and accidents in the United States were increasing annually. While the criteria for our project was defined by the University of Maryland, we had to consider the constraints presented by purchasing materials which were not provided to us as well as

sharing the resources we had with other teams. As usual though, the biggest constraint for our group was time. In order to meet all the needs of the project we had to divide parts of the project among the team and assist with other parts of the project when finished with ours.

STEM Principle for High-Priority Design Requirement #3

STEM Concept:

CCSS.ELA-LITERACY.RST.11-12.7 (Common Core ELA Standard)

Describe how STEM concept applies directly to the specifics of your design:

In order to solve the problem of a lack of effective fire detection and suppression systems, our team had to find reliable sources online which detail different types of detection and suppression systems. For background knowledge, we looked at quantitative data of fire-related deaths and accidents.

Proposed Solution to STEM Principles and Practices (E2)

STEM Principle for Design Solution #1

STEM Concept:

CCSS.MATH.CONTENT.4.MD.A.1 (Common Core 4th Grade Math: Measurement & Data)

Describe how STEM concept applies directly to the specifics of your solution:

Fourth grade measurement and data knowledge was used extensively in measuring cardboard, fabric, and other materials to the required amounts. Units of measurement had to be converted (inches to feet) in order to measure the right amount of materials needed. This was extremely crucial as the amount of resources which were available was a constraint, and so needed to be used carefully.

STEM Principle for Design Solution #2

STEM Concept:



HS-ETS1-4 Engineering Design (NGSS)

Describe how STEM concept applies directly to the specifics of your solution:

At the start of our project, all members of the group created their floor plan online on Planner 5D. Out of those designs, the best design was selected and followed through with. In order to design the actual detection/notification system, our group used the Arduino software to code the alarm. We determined how the alarm would interact with the fire so that it would beep loudly when the fire was detected. Mathematical and computational skills were used by the coder.

STEM Principle for Design Solution #3

STEM Concept:

CCSS.MATH.CONTENT.HSG.MG.A.3 (Common Core Geometry Modeling Standard)

Describe how STEM concept applies directly to the specifics of your solution:

In order to satisfy the physical requirements of creating furniture, we had to consider the constraint presented by our limited materials. Geometric skills were used to design furniture which met the required measurements and used the least amount of materials possible.

Evidence Design is Reviewed (E3)

Documentation #1:

- What format: Conversation

Our field expert (Dr. Jim Milke) talked about the connections between the design requirements and STEM principles face-to-face. Therefore, we do not have any documented evidence of our conversation.

Dr. Milke did mention that our fire alarm (detection system) works very well. It beeped



loudly as soon as the fire began, indicating that it was coded and placed well within the house. This demonstrates the use of mathematical and computational skills by the coder which allowed them to solve a real world problem.

Documentation #2:

- What format: Conversation

Our field expert (Dr. Jim Milke) talked about the connections between the design requirements and STEM principles face-to-face. Therefore, we do not have any documented evidence of our conversation. He told us that the volume of the water balloons (sprinkler system) should have been increased. There should have been more water balloons.

Dr. Milke informed us that our suppression system, which consisted of mini plastic bags filled with water, would not have worked even if the bags did not pop beforehand because there was not enough water to suppress the fire. He mentioned that it would have worked had we made more bags or made the bags bigger.

Experts with Credentials (E4)

Expert #1: Dr. Jim Milke→ professor at University of Maryland's Fire Protection Engineering Department; B.S. in Fire Protection Engineering; M.S. in Mechanical Engineering; Ph.D. in Aerospace Engineering

Verification or Corrective Response to Design Review (E5)

"_____ told us _____, which informed our decision to change/reconsider _____." (or "...which verified/reinforced our decision to _____.")

Interview with Expert #1 informs corrective response: Dr. Jim Milke told us that our detection system works well while our suppression system requires significant improvement, which



informed our decision to change the amount of water in the water balloons and add more water balloons for future projects.